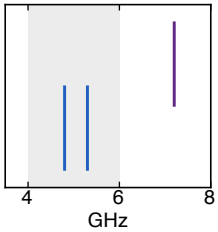
1 Specify

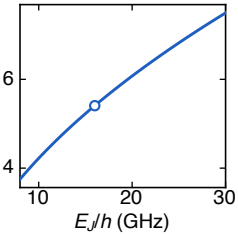
Fix the band, the readout and the lifetime goals.



You use: pen paper

2 Junction

Pick E_J for the target f_{01} at fixed E_C.



You use: pen {} code

3 Capacitances

Solve the Maxwell matrix; read E_C from its inverse.

$$E_C = \frac{e^2}{2C_\Sigma}$$

You use: EM

4 Hamiltonian

Write every term and keep three levels per mode.

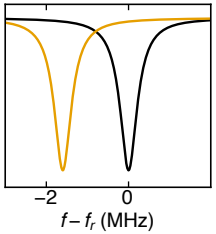
$$H/h = f_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2$$

$$+ f_r a^\dagger a + g(a^\dagger b + a b^\dagger)$$

You use: pen

5 Readout

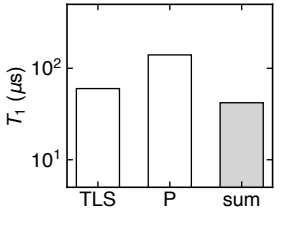
Size κ near $2|\chi|$ and check the Purcell limit.



You use: {} code VNA

6 Loss budget

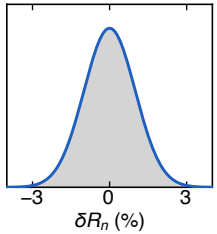
Add surface losses and Purcell decay in series.



You use: EM stats

7 Fabricate

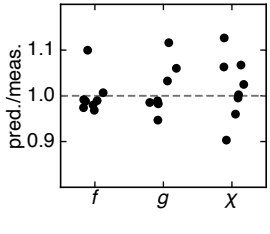
Write the mask; probe the junctions at room temperature.



You use: probe

8 Validate

Compare prediction and measurement; refit E_J.



You use: fridge ADC

Optional: tunable qubit

Replace the junction by a SQUID and budget flux noise.

Optional: package

Simulate the enclosure; keep its modes above the band.